

Contrastive Learning for Domain-Robust Text Classification

Abstract

We introduce a contrastive learning framework to improve cross-domain robustness.

This is a fictional demonstration paper, not a published research result.

The model learns transferable representations for sentiment classification.

Contrastive Learning for Domain-Robust Text Classification

1 Introduction

Text classifiers often lose accuracy when evaluated on a new domain.

We study how contrastive learning can reduce this distribution gap.

Contrastive Learning for Domain-Robust Text Classification

2 Method

We use a BERT encoder and a projection head to obtain an embedding.

Positive pairs share sentiment across domains. Negative pairs differ in sentiment.

The training objective combines classification loss and contrastive loss.

A temperature parameter controls the contrastive softmax distribution.

Contrastive Learning for Domain-Robust Text Classification

3 Experiments

Dataset: Books, Electronics, DVD (fictional demonstration data).

Preprocessing: tokenize text, then split into training and test sets.

Train ERM and domain-adversarial baselines and our contrastive model.

Evaluate all models on the same held-out test set using F1-score.

Contrastive Learning for Domain-Robust Text Classification

4 Results

Method	Books	Electronics	DVD
ERM	78.4	76.9	75.8
Domain-Adversarial	80.1	78.2	77.6
Ours	83.7	82.9	81.8

F1-score balances precision and recall; higher is better.

These numbers are fictional and must not be cited as research evidence.

Contrastive Learning for Domain-Robust Text Classification

5 Conclusion

Contrastive representations may improve transfer across text domains.

This demo illustrates the reading workflow only.

Limitations: no confidence intervals, no real dataset or reproducibility claim.